

1. If R is the region bounded by  $x = 0, y = 0, z = 0, z = 1$  and the cylinder  $x^2 + y^2 = 1$  evaluate  $\iiint xyz dx dy dz$  by changing to cylindrical coordinates. Ans:  $\frac{1}{16}$
2. Evaluate  $\iiint (x^2 + y^2) dx dy dz$  over the region bounded by the paraboloid  $x^2 + y^2 = 3z$  and the plane  $z = 3$  ans:  $\frac{81\pi}{2}$
3. Evaluate  $\iiint \frac{dx dy dz}{\sqrt{x^2 + y^2 + z^2}}$  over the region bounded by the sphere  $x^2 + y^2 + z^2 = a^2$  and  $x^2 + y^2 + z^2 = b^2$   $a > b > 0$  ans:  $2\pi(a^2 - b^2)$

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